

Chua Heng Li

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Portfolio: <https://chuahengli.github.io/Portfolio-Site/>

EDUCATION

Singapore University of Technology and Design (SUTD)

Sep 24 - Present

Bachelor of Engineering in Engineering Product Development (Electrical Track)

Singapore

- cGPA: 4.2/5.0 | SUTD Global Merit Scholarship
- Relevant Coursework: Circuits & Electronics, Computational & Data-Driven Engineering, EM & Applications, Systems & Controls
- APEX Club Vice President; Spearheaded a cross-university SG60 kayaking event at PassionWave@Sembawang, coordinating participation from SMU, NUS and NTU

Jurong Pioneer Junior College (JPJC)

Jan 20 - Dec 21

A' Levels

Singapore

- RP: 86.25 | H3 Physics (Semiconductor Devices) with NTU
- Quartermaster of Guitar Ensemble; Represented JPJC at Singapore Youth Festival (SYF), earning Certificate of Commendation; managed instrument inventory and weekly rehearsals

SKILLS

- **Programming Languages:** Python, C, C++
- **Libraries & Frameworks:** pandas, NumPy, Plotly, Streamlit, scikit-learn
- **Hardware & Design Tools:** Arduino, KiCad (PCB Design), AutoCAD, Fusion360, CST Microwave Studio
- **Development Tools:** Git, GitHub
- **Languages:** English, Mandarin

PROJECTS

Stock Portfolio Tracker

Nov 25 - Feb 26

Personal Project

- Engineered a Python-based automated data pipeline integrating Moomoo's OpenD API to retrieve and process daily portfolio data
- Designed a SQLite database schema to persist and query historical portfolio metrics and transaction records
- Built an interactive Streamlit dashboard with Plotly visualizations, benchmarking portfolio returns against major world indices via Yahoo Finance API
- Packaged the pipeline as a one-command daily refresh so portfolio data updates and re-benchmarks against world indices at each market close

3D Printed Bird Deterrent Device

Jan 25 - May 25

SUTD Course Project

- Applied design thinking to develop a sensor-triggered bird deterrent prototype addressing contamination in an open-air canteen
- Programmed an Arduino with PIR motion sensors to activate a 3D-printed gear mechanism upon detecting bird movement
- Fabricated the prototype through woodworking and iterative design; awarded the Sustainability Practice Prize for sustainable design and material use

Concentric Circular Antenna Simulation

Jul 26

SUTD Electromagnetism & Applications | CST Microwave Studio

- Designed and simulated a concentric circular antenna for the 2.45 GHz ISM band in CST (S11 / return loss)
- Ran parametric analysis through the concentric circular stage before teammate handover for integration

Bed Assisted Device | ST Engineering 'For Good'

Jan 26 - May 26

SUTD 3D Design Project | Monte Carlo simulation, DC motor sizing

- Modeled motor torque via a 1M-run Monte Carlo simulation (user mass, friction, shaft radius): 1.026 Nm to assist 95% of users
- Specified a dual 12V DC gear motor w/ encoder, validating rated vs stall torque/current for HDB corridor-step assist

WORK EXPERIENCE

UOB Kay Hian

Mar 24 - Sep 24

Operations Assistant

Singapore

- Processed and reconciled dividend announcement data for client accounts using Microsoft Excel across multiple securities and corporate-action events
- Maintained structured client records and coordinated timely, accurate client notifications with the operations team